

Research papers

Seasonal operation planning of hydrogen-enabled multi-energy microgrids through multistage stochastic programming[☆]

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ABSTRACT

As a carbon-free storage medium, hydrogen has advantages in large-scale and long-term energy shifting, thereby mitigating the seasonal imbalance between energy supply and demands. By integrating hydrogen, electricity, heating and cooling, the hydrogen-enabled multi-energy microgrid (HMM) provides a desirable test bed for decarbonizing the energy and power systems. In this paper, we study the networked HMMs operation planning (NHOP) that optimizes the multi-timescale synergy of seasonal and short-term energy storage. To hedge against the complex demand-supply uncertainties (e.g., seasonal fluctuation and hourly variation of renewable power generation and energy demands), the NHOP problem is recast as a multistage stochastic mixed-integer program (MS-MIP). Moreover, to overcome the computational challenges, a nested decomposition algorithm based on stochastic dual dynamic integer programming (SDDiP) is tailored and implemented. Case studies on a 33-bus test network with multiple HMMs demonstrate the economic benefits of our operation planning strategy. The proposed NHOP model can well capture the seasonal and intra-day dynamics of multi-type storage operation, which helps improve the cost-benefits under versatile practical situations. Also, the customized SDDiP algorithm shows a strong scalable capacity for efficiently computing large-scale MS-MIPs.

1. Introduction

Global climate change poses one of the greatest challenges to the sustainable development of humanity. The target of controlling global warming demands for the efforts of multiple social sectors to achieve the carbon neutrality in the long-run [1]. The energy sectors, which have contributed to nearly 40% CO₂ emissions [2], are particularly urgent for a revolutionary transition to clean and low-carbon operation patterns. Among the alternatives to carbon-intensive fossil fuels, hydrogen, which bears the merits of diverse sources, zero carbon emissions and high energy density [3], has tremendous potential for a wide range of applications in energy and power systems [4,5], transportation [6,7], and industrial sectors [8]. Notably, integrating hydrogen with renewable energy sources (RESs), e.g., photovoltaic arrays (PVs) and wind turbines (WTs), may contribute to fostering a sustainable energy future. In this way, renewable electricity can be effectively stored by converting to hydrogen, enabling its subsequent on-site utilization

or long-distance transport to mitigate the short-term (e.g., daily and weekly) as well as long-term (e.g., monthly and seasonal) imbalances in energy supply and demands without carbon emissions [3].

With the recent advances in hydrogen production and storage technologies, developing the hydrogen-enabled multi-energy microgrids (HMMs) becomes a viable pathway for revolutionizing the energy supply infrastructure. By converting the excessive generation of RESs to “green” hydrogen through water electrolyzers (ELZs), the potential renewable curtailment can be largely reduced. Also, the produced hydrogen would be converted back to feed the electrical power system via fuel cells (FCs) during the periods that RES outputs are inadequate, along with the generation of thermal energy [9]. For a typical proton exchange membrane FC (PEMFC) stack, its power and thermal generation efficiencies are 35%–39% and 55%, respectively [10]. The thermal energy can be further collected to satisfy local heating and cooling demands, thereby establishing a hydrogen-enabled combined cooling,

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Nomenclature**Abbreviation**

AC	Absorption chiller
BS	Battery storage
COP	Coefficient of performance
CWS	Chilled water storage
ELZ	Electrolyzer
FC	Fuel cell
H-CCHP	Hydrogen-enabled combined cooling, heating and power
HMM	Hydrogen-enabled multi-energy microgrid
HS	Hydrogen storage
HWS	Hot water storage
LV	Low-voltage
MS-MIP	Multistage stochastic mixed-integer program
MV	Medium-voltage
NHOP	Networked HMMs operation planning
OPEX	Operating expense
PDN	Power distribution network
PV	Photovoltaic array
RES	Renewable energy source
SDDiP	Stochastic dual dynamic integer programming
SHS	Seasonal hydrogen storage
SOC	State of charge
WT	Wind turbine

Sets and Indices

(t, d, h)	Time index of operating period h in typical day d of season t
Λ	Set of buses with HSS deployment
$\mathcal{L}/(i, j)$	Set/index of lines of PDN
$\mathcal{N}/i, j, l$	Set/index of buses of PDN
Ω_{res}	Set of RESs
$\Theta(i)$	Set of children buses of bus i
D	Number of typical days of a season
d	Index of typical days of season t
H	Number of operating periods in a typical day
h	Index of operating periods in typical day
k	Index of candidate facilities
t	Index of seasons

Parameters

$\beta_{bs}^{\max}$	Maximum depth of discharge of battery storage unit
Δ_h	Duration of operating period h
$\delta_{k,i}^{t,d,h}$	Capacity factor of RES k restricted by natural resources at bus i at (t, d, h)
η_{k+}/η_{k-}	Charge/discharge efficiency of storage k
$t_{es}/t_{hs}/t_{cs}$	Penalty cost factor of electrical/heating/cooling load shedding
ψ_k	Loss factor of component k
σ	Scale factor from typical days to a season

κ_{ec}	Penalty cost factor of electricity curtailment
$\varphi_k^{\max}/\varphi_k^{\min}$	Max-/minimum power factor angle of inverter connected with component k
$\varphi_{pl,i}^{t,d,h}$	Power factor angle of electrical demands at bus i at (t, d, h)
$\rho_{imp}^{t,d,h}$	Utility prices for power procurement at (t, d, h)
ζ_k	Degradation price of component k
$P_{k,i}/C_{k,i}$	Power/energy capacity of component k at bus i
$PD_i^{t,d,h}/HD_i^{t,d,h}/CD_i^{t,d,h}$	Electrical/heating/cooling demands at bus i at (t, d, h)
r_{ij}/x_{ij}	Resistance/reactance of line (i, j)
$S_{ij}^{\max}$	Capacity limit of line (i, j)
$S_{lv,i}^{\max}$	Capacity limit of LV substation at bus i
$S_{mv}^{\max}$	Capacity limit of MV substation
T	Number of seasons in operation horizon
$U_i^{\max}/U_i^{\min}$	Voltage magnitude upper/lower bound at bus i

Decision Variables

$e_{bs,i}^{t,d,h}$	SOC of BSs at bus i at (t, d, h)
$f_{p_{ij}}^{t,d,h}/f_{q_{ij}}^{t,d,h}$	Active/reactive power flow on line (i, j) at (t, d, h)
$g_{k+,i}^{t,d,h}/g_{k-,i}^{t,d,h}$	Hydrogen inflow/outflow of HSs/SHSs at bus i at (t, d, h)
$g_{k,i}^{t,d,h}$	SOC of HSs/SHSs at bus i at (t, d, h)
$m_{ac,i}^{t,d,h}/n_{ac,i}^{t,d,h}$	Heating inflow/cooling outflow of ACs at bus i at (t, d, h)
$m_{cws,i}^{t,d,h}$	SOC of CWSs at bus i at (t, d, h)
$m_{hws+,i}^{t,d,h}/m_{hws-,i}^{t,d,h}$	Heating inflow/outflow of HWSs at bus i at (t, d, h)
$m_{hws,i}^{t,d,h}$	SOC of HWSs at bus i at (t, d, h)
$n_{cws+,i}^{t,d,h}/n_{cws-,i}^{t,d,h}$	Cooling inflow/outflow of CWSs at bus i at (t, d, h)
$p_{bs+,i}^{t,d,h}/p_{bs-,i}^{t,d,h}$	Charging/discharging power of BSs at bus i at (t, d, h)
$p_{elz,i}^{t,d,h}/g_{elz,i}^{t,d,h}$	Power input/hydrogen outflow of ELZs at bus i at (t, d, h)
$p_{fc,i}^{t,d,h}/m_{fc,i}^{t,d,h}/g_{fc,i}^{t,d,h}$	Power/heating output/hydrogen inflow of FCs at bus i at (t, d, h)
$p_{k,i}^{t,d,h}/q_{k,i}^{t,d,h}$	Active/reactive power output of component k at bus i at (t, d, h)
$p_{lv,i}^{t,d,h}/q_{lv,i}^{t,d,h}$	Active/reactive power via lv substation at bus i at (t, d, h)
$p_{mv}^{t,d,h}/q_{mv}^{t,d,h}$	Active/reactive power via mv substation at (t, d, h)
$pl_i^{t,d,h}/hl_i^{t,d,h}/cl_i^{t,d,h}$	Satisfied electrical/heating/cooling demands at bus i at (t, d, h)
$v_i^{t,d,h}$	Squared magnitudes of voltage at bus i at (t, d, h)
$z_{k,i}^{t,d,h}$	Binary, 1 if storage k at bus i charges at (t, d, h) , and 0 otherwise
$z_{shs,i}^t$	Binary, 1 if SHS at bus i charges at t , and 0 otherwise

heating and power (H-CCHP) system [11]. Note that the aforementioned energy supply procedures is with zero carbon and pollutants emission. Additionally, when connected to the power distribution network (PDN), HMMs can actively participate in the demand response

(DR) programs or power regulation markets to improve the economic benefits [7,12].

However, at the current stage, the hydrogen production through electrolysis could be very costly [13]. It is critical to optimize the operation of HMMs for enhancing the economic efficiency. To date, the operational optimization of HMMs has been extensively studied in many existing literature. Ref. [14] investigated the optimal operation of hydrogen-based integrated systems, coordinating hydrogen, electricity, cooling and heating. This study indicated that the synergy of hydrogen and water storage has brought an 80% reduction in energy costs, while achieving nearly 100% RES accommodation. Due to the low efficiency of large-scale water electrolysis-based hydrogen production, an optimal scheduling strategy for hydrogen-based microgrids was proposed in Ref. [15], resulting in a 12.5% enhancement of ELZs' utilization rates. In Ref. [16], a market-oriented multi-timescale energy management method for HMMs, comprising day-ahead and real-time economic dispatch, was studied, which could reduce the average diurnal operational costs by 37.08%. Considering the distinguished preferences of different operators, Ref. [17] proposed a multi-objective scheduling approach for integrated power-hydrogen systems based on distributed ξ -constraint optimization. This method was to optimize the main objective function while ensuring the interests of other objective terms. To facilitate the resilient operations against extreme events, e.g., natural disaster and malicious attacks, Ref. [18] developed a risk-averse rolling-horizon scheduling method for HMMs. As revealed by the numerical results, this method could potentially delay the occurrence time of power supply interruption, and improve the load survivability under disastrous occasions. Similarly, Ref. [19] presented a proactive scheduling method for resilience-oriented operation of hydrogen systems. Moreover, considering the coordination of energy and transportation sectors, Ref. [20] formulated an economical scheduling method for microgrids integrating hydrogen fueling stations, facilitating optimal on-site hydrogen generation and refueling services for hydrogen vehicles (HVs).

We have noticed that, in these studies, the hydrogen storage is usually utilized for intra-day or real-time energy regulation, which demonstrates a similar functionality to ordinary storage types (e.g., batteries, supercapacitors or flywheel storage). Compared to conventional storage facilities, however, the hydrogen storage is with a higher energy density, which could be more suitable to perform the large-scale and long-term energy shifting (e.g., the seasonal storage through underground salt caverns) [21]. Such technical advantage has significant practical meanings for its application to handle the volatile and random RES generation, which may follow an evident seasonal rhythm [22]. For instance, in Northern China, the peaks of solar generation generally occur in Summer, while the wind power resources could be more abundant during Spring and Winter months. On the other hand, the seasonal pattern of multi-energy demands is also non-negligible [11]. For instance, cooling and heating demands could be concentratedly consumed in Summer and Winter respectively, which happens to be consistent with the peaking periods of power consumption. To keep track with the seasonal fluctuation of energy supply and demands, we believe that the seasonal hydrogen storage (SHS) needs to be properly configured and optimally operated.

A few recent studies have concerned the planning and scheduling of multi-energy systems equipping with SHS devices, e.g., in Refs. [21, 23–26]. Ref. [21] presented a mixed integer program (MIP) formulation to support the system design incorporating SHS. It indicated that hydrogen storage operates in a seasonal pattern only when a high storage capacity is available, which enabled smooth charging and discharging. Ref. [23] showed that the SHS could further improve the economical benefits of hydrogen-based integrated energy systems, by addressing the randomness of RES outputs. Ref. [24] proposed a risk-constrained stochastic planning method for off-grid microgrids under economics and resilience considerations. In this study, the leveled system cost could be reduced by 0.0746 \$/kWh through proper investment decisions on SHS and short-term energy storage. Ref. [25]

studied the energy management strategies of an off-grid microgrid equipping with RESs, SHS and short-term battery storage. Through the generated strategy, SHS could be fully exploited to maintain the long-term energy balance, reducing the load shedding and renewable energy curtailment. In Ref. [26], an optimal scheduling method based on Deep Deterministic Policy Gradient (DDPG) was developed for an integrated energy system with SHS, which improved the energy utilization efficiency, and mitigated the dependence on external energy networks. In the existing publications, however, there lacks an in-depth exploration on the complex and multi-timescale interaction among SHS and various short-term storage devices. So it could be difficult to induce a long-term economical operation strategy for networked HMMs to realize the synergy of multiscale energy storage. Moreover, the current uncertainty models are essentially static, which cannot capture the dynamic storage scheduling decisions against the seasonal randomness of RES generation and energy demands.

To address these challenges, this paper studied the long-term operational optimization of HMMs integrating RES units, SHS, short-term energy storage (i.e., the daily hydrogen storage, battery banks and thermal storage vessels) and H-CCHP systems. For the first time, the networked HMMs operation planning (NHOP) problem is formulated as a multistage program for characterizing the seasonal and diurnal synergy of multiscale energy storage. Then, to handle the complex supply-demand uncertainties, a stochastic extension of NHOP is presented. By leveraging the specific structure of scenario tree to discretely represent the stochastic processes (including seasonal large-scale fluctuation and hourly small-scale variation of RES outputs as well as the electrical, heating and cooling demands), we derive a multistage stochastic mixed-integer program (MS-MIP) model. Moreover, based on stochastic dual dynamic integer programming (SDDiP), a nested decomposition algorithm is developed and customized to reduce the computational burden for our stochastic formulation.

Compared to the current studies, the major contributions of this paper can be summarized as three-fold:

1. A seasonal NHOP formulation is proposed, where the dynamic interaction of short- and long-term energy storage, as well as various energy conversion units are analytically modeled.
2. An MS-MIP extension for NHOP is developed to control the operational risks of HMMs associating with the multiscale supply-demand uncertainties. Also, an SDDiP-based nested decomposition algorithm is designed and implemented to efficiently solve this MS-MIP model.
3. Numerical studies on a 33-bus networked microgrids system under real-world RES data demonstrate the economical benefits of the proposed method. Through the coordinated scheduling of SHS and short-term energy storage, a higher operational flexibility can be achieved to reduce the risk cost.

The remainder of this paper is organized as follows. Section 2 proposes the NHOP formulation and its stochastic extension. Section 3 developed the nested decomposition algorithm based on SDDiP. Then, we perform case studies in Section 4 to demonstrate the effectiveness of our NHOP model and SDDiP algorithm. Finally, conclusions are attained and discussed in Section 5.

2. Problem formulation

We study the NHOP problem in an industry park. As exhibited in Fig. 1, driven by RESs, i.e., PVs and WTs, an HMM produces “green” hydrogen through ELZs. Then, by utilizing FCs, HMM could convert hydrogen to electricity, and produce heat simultaneously. Integrating FCs, heat recovery units (HRs) and absorption chillers (ACs), an H-CCHP system is built, so as to satisfy the local electricity, cooling and heating demands. The whole procedure mentioned above should be noted for its zero-carbon emissions. In order to cope with the short-term mismatch between supply and demand sides, battery storage (BS)

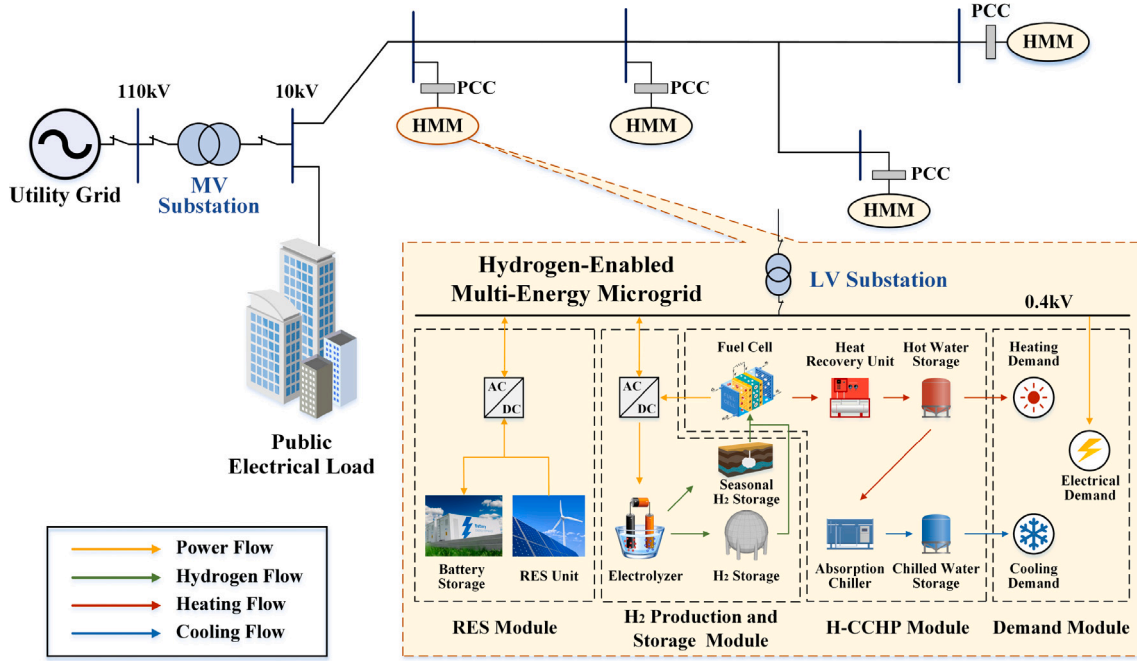


Fig. 1. Illustrative structure of networked hydrogen-enabled multi-energy microgrids.

units, hydrogen storage (HS) units, hot water storage (HWS) units and chilled water storage (CWS) units are incorporated to store electricity, hydrogen, heating and cooling, respectively. Besides, the SHS units are also considered to further mitigate the long-term supply-demand imbalance.

HMMs are connected to the supporting PDN, which possesses a spanning-tree topology. The sets of PDN buses and lines are denoted as $\mathcal{N}$ and $\mathcal{L}$, respectively. We set the medium-voltage (MV) substation at bus 0, while the corresponding subset of buses with HMM deployment is denoted as Λ . The operation horizon is composed of T seasons, in which each season is modeled by D successive typical days. Also, each typical day is discretized by H operating periods. In the following of this section, the basic deterministic formulation of NHOP is provided.

2.1. Objective function

The objective of NHOP is to minimize the total operation expenses (Φ_{opex}) upon the operation horizon, which can be expressed as:

$$\Phi_{opex} = \min \sum_{t=1}^T \Phi_{opex}^t \quad (1)$$

where Φ_{opex}^t represents OPEXs during season t . In (2), Φ_{opex}^t includes electricity purchase costs (line 1), degradation costs of BSs, ELZs and FCs (line 2 and 3), and penalty costs of RES curtailment (line 4) and load shedding (line 5 and 6). Time-related factor $\delta_{k,i}^{t,d,h}$ is introduced to describe the availability of renewable natural resources, i.e., wind and solar. Also, the electricity procurement price, $\rho_{imp}^{t,d,h}$, is in the time-of-use (TOU) mode.

$$\begin{aligned} \Phi_{opex}^t = & \sigma \sum_{d \in \mathcal{D}^t} \sum_{h \in \mathcal{H}^{t,d}} \left[\sum_{i \in \Lambda} \left(\rho_{imp}^{t,d,h} p_{mv}^{t,d,h} \Delta_h + \right. \right. \\ & + \sigma \sum_{d \in \mathcal{D}^t} \sum_{h \in \mathcal{H}^{t,d}} \left[\sum_{i \in \Lambda} \left(\frac{1}{2} \zeta_{bs} \left(p_{bs+,i}^{t,d,h} + p_{bs-,i}^{t,d,h} \right) \Delta_h + \right. \right. \\ & \left. \left. \sum_{i \in \Lambda} \left(\zeta_{elz} p_{elz,i}^{t,d,h} \Delta_h + \zeta_{fc} p_{fc,i}^{t,d,h} \Delta_h \right) \right] \right. \\ & + \sigma \sum_{d \in \mathcal{D}^t} \sum_{h \in \mathcal{H}^{t,d}} \left[\sum_{i \in \Lambda} \sum_{k \in \Omega_{res}} x_{ec} \left(\delta_{k,i}^{t,d,h} P_{k,i} - p_{k,i}^{t,d,h} \right) \Delta_h \right] \\ & \left. + \sigma \sum_{d \in \mathcal{D}^t} \sum_{h \in \mathcal{H}^{t,d}} \left[\sum_{i \in \mathcal{N}} l_{es} \left(P D_i^{t,d,h} - p_i^{t,d,h} \right) \Delta_h + \right. \right. \end{aligned} \quad (2)$$

$$\sum_{i \in \Lambda} l_{hs} \left(H D_i^{t,d,h} - h_i^{t,d,h} \right) \Delta_h + \sum_{i \in \Lambda} l_{cs} \left(C D_i^{t,d,h} - c_i^{t,d,h} \right) \Delta_h \Big]$$

2.2. Long-term operation constraints

According to each season $t = 1, \dots, T$, the operational constraints of NHOP are summarized in this subsection.

2.2.1. Renewable energy source

The constraints of RESs (i.e., WTs and PVs) are given in (3) and (4). Note that the availability of RES outputs depends on natural resources ($\delta_{k,i}^{t,d,h}$).

$$0 \leq p_{k,i}^{t,d,h} \leq \delta_{k,i}^{t,d,h} P_{k,i}, \quad \forall k \in \Omega_{res}, \forall i \in \Lambda, \forall d, \forall h \quad (3)$$

$$p_{k,i}^{t,d,h} \tan \varphi_k^{\min} \leq q_{k,i}^{t,d,h} \leq p_{k,i}^{t,d,h} \tan \varphi_k^{\max}, \quad \forall k \in \Omega_{res}, \forall i \in \Lambda, \forall d, \forall h \quad (4)$$

2.2.2. Battery storage

Operation of BSs is confined by (5)–(9). $P_{bs,i}$ and $E_{bs,i}$ represent power and energy capacities of BS, respectively. By using 0–1 variable $z_{bs,i}^{t,d,h}$, the charging and discharging power are imposed by (5) and (6). (7) presents the dynamic equation, reflecting the relationship of state-of-charge (SOC) between adjacent periods, and SOC is limited by (8). Considering the charging–discharging cycle, (9) confirms the net charging capacities to be zero during one-day operation.

$$0 \leq p_{bs+,i}^{t,d,h} \leq P_{bs,i} z_{bs,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (5)$$

$$0 \leq p_{bs-,i}^{t,d,h} \leq P_{bs,i} (1 - z_{bs,i}^{t,d,h}), \quad \forall i \in \Lambda, \forall d, \forall h \quad (6)$$

$$e_{bs,i}^{t,d,h+1} = e_{bs,i}^{t,d,h} + \left(p_{bs+,i}^{t,d,h} \eta_{bs+} - p_{bs-,i}^{t,d,h} / \eta_{bs-} \right) \Delta_h, \quad \forall i \in \Lambda, \forall d, \forall h \quad (7)$$

$$(1 - \beta_{bs}^{\max}) C_{bs,i} \leq e_{bs,i}^{t,d,h} \leq C_{bs,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (8)$$

$$\sum_{h=1}^H \left(p_{bs+,i}^{t,d,h} \eta_{bs+} - p_{bs-,i}^{t,d,h} / \eta_{bs-} \right) \Delta_h = 0, \quad \forall i \in \Lambda, \forall d \quad (9)$$

2.2.3. Electrolyzers

ELZ is the device for electricity-hydrogen conversion. (10) and (11) show the constraints of hydrogen production through water electrolysis. χ_{p-g} is the coefficient of performance (COP) of ELZ for electricity-hydrogen conversion.

$$g_{elz,i}^{t,d,h} = \chi_{p-g} p_{elz,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (10)$$

$$0 \leq p_{elz,i}^{t,d,h} \leq P_{elz,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (11)$$

2.2.4. Fuel cell and heat recovery unit

FC is the core component of the H-CCHP system. (12) and (13) exhibit the electrical power generated by FC. (14) calculates the captured and recovered residual heating in the power generation procedure. χ_{g-e} and χ_{g-h} are electrical and heating COPs of FC, respectively. Besides, the electrical power output is restricted by FC's capacity, as shown in (15).

$$p_{fc,i}^{t,d,h} = \chi_{g-e} g_{fc,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (12)$$

$$p_{fc,i}^{t,d,h} \tan \varphi_{fc}^{\min} \leq q_{fc,i}^{t,d,h} \leq p_{fc,i}^{t,d,h} \tan \varphi_{fc}^{\max}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (13)$$

$$m_{fc,i}^{t,d,h} = \chi_{g-h} g_{fc,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (14)$$

$$0 \leq p_{fc,i}^{t,d,h} \leq P_{fc,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (15)$$

2.2.5. Absorption chiller

Through AC, the heating energy generated by FC can be transformed to cooling. (16) and (17) specify the cooling energy produced by AC. χ_{h-c} is COP of AC.

$$n_{ac,i}^{t,d,h} = \chi_{h-c} m_{ac,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (16)$$

$$0 \leq n_{ac,i}^{t,d,h} \leq P_{ac,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (17)$$

2.2.6. Short-term hydrogen, heating and cooling storage

Eqs. (18), (23) and (26) represent the storage dynamics of short-term HS, HWS and CWS, respectively. (19) and (20) restrict the charging and discharging state of HT by a 0–1 variable $z_{hs,i}^{t,d,h}$. Similar to (8), Eqs. (21), (24) and (27) describe the constraints associated with the daily storage cycle. (22), (25) and (28) provide the limitations corresponding to HS, HES and CWS capacities.

$$g_{hs,i}^{t,d,h+1} = (1 - \psi_{hs}) g_{hs,i}^{t,d,h} + (g_{hs+,i}^{t,d,h} \eta_{hs+} - g_{hs-,i}^{t,d,h} / \eta_{hs-}) \Delta_h, \quad \forall i \in \Lambda, \forall d, \forall h \quad (18)$$

$$0 \leq g_{hs+,i}^{t,d,h} \leq C_{hs,i} z_{hs,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (19)$$

$$0 \leq g_{hs-,i}^{t,d,h} \leq C_{hs,i} (1 - z_{hs,i}^{t,d,h}), \quad \forall i \in \Lambda, \forall d, \forall h \quad (20)$$

$$\sum_{h \in \mathcal{H}^{t,d}} (g_{hs+,i}^{t,d,h} \eta_{hs+} - g_{hs-,i}^{t,d,h} / \eta_{hs-}) = 0, \quad \forall i \in \Lambda, \forall d, \forall h \quad (21)$$

$$0 \leq g_{hs,i}^{t,d,h} \leq C_{hs,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (22)$$

$$m_{hws,i}^{t,d,h+1} = (1 - \psi_{hws}) m_{hws,i}^{t,d,h} + (m_{hws+,i}^{t,d,h} \eta_{hws+} - m_{hws-,i}^{t,d,h} / \eta_{hws-}) \Delta_h, \quad \forall i \in \Lambda, \forall d, \forall h \quad (23)$$

$$\sum_{h \in \mathcal{H}^{t,d}} (m_{hws+,i}^{t,d,h} \eta_{hws+} - m_{hws-,i}^{t,d,h} / \eta_{hws-}) \Delta_h = 0, \quad \forall i \in \Lambda, \forall d, \forall h \quad (24)$$

$$0 \leq m_{hws,i}^{t,d,h} \leq C_{hws,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (25)$$

$$n_{cws,i}^{t,d,h+1} = (1 - \psi_{cws}) n_{cws,i}^{t,d,h} + (n_{cws+,i}^{t,d,h} \eta_{cws+} - n_{cws-,i}^{t,d,h} / \eta_{cws-}) \Delta_h, \quad \forall i \in \Lambda, \forall d, \forall h \quad (26)$$

$$\sum_{h \in \mathcal{H}^{t,d}} (n_{cws+,i}^{t,d,h} \eta_{cws+} - n_{cws-,i}^{t,d,h} / \eta_{cws-}) \Delta_h = 0, \quad \forall i \in \Lambda, \forall d, \forall h \quad (27)$$

$$0 \leq n_{cws,i}^{t,d,h} \leq C_{cws,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (28)$$

2.2.7. Seasonal hydrogen storage

Eqs. (29)–(35) provide the constraints for long-term SHS. (29) is the intra-day dynamic equation. (30) confirms the continuity of hydrogen SOC between two typical days. Since we only consider D typical days in each season, we introduce $g_{shs,i}^{t,end,H+1}$ to denote the actual SOC after the last day of season t , which equals to the initial SOC of next season $t+1$. (31) and (32) show the relationship among $g_{shs,i}^{t,end,H+1}$, $g_{shs,i}^{t,1,1}$, $g_{shs,i}^{t,D,H+1}$ and $g_{shs,i}^{t+1,1,1}$. Π is the scale factor for D typical days to one season. (33) is the capacity constraint for hydrogen SOC. With a seasonal 0–1 state

variable $z_{shs,i}^t$, (34) and (35) impose the charging and discharging of SHS units.

$$g_{shs,i}^{t,d,h+1} = (1 - \psi_{shs}) g_{shs,i}^{t,d,h} + (g_{shs+,i}^{t,d,h} \eta_{shs+} - g_{shs-,i}^{t,d,h} / \eta_{shs-}) \Delta_h, \quad \forall i \in \Lambda, \forall d, \forall h \quad (29)$$

$$g_{shs,i}^{t,d,1} = g_{shs,i}^{t,d-1,H+1}, \quad \forall i \in \Lambda, \forall d \quad (30)$$

$$g_{shs,i}^{t,end,H+1} = g_{shs,i}^{t,1,1} + \Pi (g_{shs,i}^{t,D,H+1} - g_{shs,i}^{t,1,1}), \quad \forall i \in \Lambda \quad (31)$$

$$g_{shs,i}^{t+1,1,1} = g_{shs,i}^{t,end,H+1}, \quad \forall i \in \Lambda \quad (32)$$

$$0 \leq g_{shs,i}^{t,d,h}, g_{shs,i}^{t,end,H+1} \leq C_{shs,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (33)$$

$$0 \leq g_{shs+,i}^{t,d,h} \leq z_{shs,i}^t C_{shs,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (34)$$

$$0 \leq g_{shs-,i}^{t,d,h} \leq (1 - z_{shs,i}^t) C_{shs,i}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (35)$$

2.2.8. Electrical, heating and cooling demands

Considering the mechanism of load shedding, the satisfied electrical load ($pl_i^{t,d,h}$), heating load ($hl_i^{t,d,h}$) and cooling load ($cl_i^{t,d,h}$) are imposed in (36)–(38), respectively. In them, $PD_i^{t,d,h}$, $HD_i^{t,d,h}$ and $CD_i^{t,d,h}$ give the actual electrical, heating and cooling demands of the local area.

$$0 \leq pl_i^{t,d,h} \leq PD_i^{t,d,h}, \quad \forall i \in \mathcal{N}, \forall d, \forall h \quad (36)$$

$$0 \leq hl_i^{t,d,h} \leq HD_i^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (37)$$

$$0 \leq cl_i^{t,d,h} \leq CD_i^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (38)$$

2.2.9. System-level coupling constraints

Eqs. (39) and (40) represent the active and reactive power balances of an HMM. (41) limits the power transaction through the low-voltage (LV) substations, which connect to PDN. Moreover, (42)–(47) provide the coupling equations for hydrogen, heating and cooling energy components.

$$\sum_{k \in \Omega_{res}} p_{k,i}^{t,d,h} + (p_{bs-,i}^{t,d,h} - p_{bs+,i}^{t,d,h}) + p_{fc,i}^{t,d,h} + p_{lv,i}^{t,d,h} = pl_i^{t,d,h} + p_{elz,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (39)$$

$$\sum_{k \in \Omega_{res}} q_{k,i}^{t,d,h} + q_{fc,i}^{t,d,h} + q_{lv,i}^{t,d,h} = pl_i^{t,d,h} \tan \varphi_{pl,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (40)$$

$$(p_{lv,i}^{t,d,h})^2 + (q_{lv,i}^{t,d,h})^2 \leq (S_{lv,i}^{\max})^2, \quad \forall i \in \Lambda, \forall d, \forall h \quad (41)$$

$$g_{hs+,i}^{t,d,h} + g_{shs+,i}^{t,d,h} = g_{elz,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (42)$$

$$g_{hs-,i}^{t,d,h} + g_{shs-,i}^{t,d,h} = g_{fc,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (43)$$

$$m_{hws+,i}^{t,d,h} = m_{fc,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (44)$$

$$m_{hws-,i}^{t,d,h} = m_{ac,i}^{t,d,h} + hl_i^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (45)$$

$$n_{cws+,i}^{t,d,h} = n_{ac,i}^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (46)$$

$$n_{cws-,i}^{t,d,h} = cl_i^{t,d,h}, \quad \forall i \in \Lambda, \forall d, \forall h \quad (47)$$

Remark 1. Note that (41) holds a nonlinear form due to the quadratic terms $(p_{lv,i}^{t,d,h})^2$ and $(q_{lv,i}^{t,d,h})^2$, which may pose computational challenges in practice. Following the strategy mentioned in [27], (41) can be approximately replaced by the linear constraints as below:

$$\begin{aligned} -S_{lv,i}^{\max} &\leq p_{lv,i}^{t,d,h}, q_{lv,i}^{t,d,h} \leq S_{lv,i}^{\max}, \quad \forall i \in \Lambda, \forall d, \forall h \\ -\sqrt{2} S_{lv,i}^{\max} &\leq p_{lv,i}^{t,d,h} + q_{lv,i}^{t,d,h} \leq \sqrt{2} S_{lv,i}^{\max}, \quad \forall i \in \Lambda, \forall d, \forall h \\ -\sqrt{2} S_{lv,i}^{\max} &\leq p_{lv,i}^{t,d,h} - q_{lv,i}^{t,d,h} \leq \sqrt{2} S_{lv,i}^{\max}, \quad \forall i \in \Lambda, \forall d, \forall h \end{aligned}$$

2.2.10. Constraints of power distribution network

We apply a linearized *Disflow* model [28] to formulate the operation of PDN, as described in (48)–(56). Subset $\mathcal{N}^+$ is defined as $\mathcal{N} \setminus 0 =$

$\{1, 2, \dots\}$. Capacity constraints (52) and (56) can also be linearized by using the method in Remark 1.

$$f p_{ji}^{t,d,h} - \sum_{l \in \Theta(i)} f p_{il}^{t,d,h} = \begin{cases} p_{lv,i}^{t,d,h}, & \forall i \in \Lambda, \forall d, \forall h \\ p_{li}^{t,d,h}, & \forall i \in \mathcal{N}^+ \setminus \Lambda, \forall d, \forall h \end{cases} \quad (48)$$

$$f q_{ji}^{t,d,h} - \sum_{l \in \Theta(i)} f q_{il}^{t,d,h} = \begin{cases} q_{lv,i}^{t,d,h}, & \forall i \in \Lambda, \forall d, \forall h \\ p_{li}^{t,d,h} \tan \varphi_{pl,i}^{t,d,h}, & \forall i \in \mathcal{N}^+ \setminus \Lambda, \forall d, \forall h \end{cases} \quad (49)$$

$$\sum_{l \in \Theta(0)} f p_{0l}^{t,d,h} = p_{mv}^{t,d,h} \geq 0, \quad \forall d, \forall h \quad (50)$$

$$\sum_{l \in \Theta(0)} f q_{0l}^{t,d,h} = q_{mv}^{t,d,h} \geq 0, \quad \forall d, \forall h \quad (51)$$

$$\left(f p_{ij}^{t,d,h}\right)^2 + \left(f q_{ij}^{t,d,h}\right)^2 \leq \left(S_{ij}^{\max}\right)^2, \quad \forall (i, j) \in \mathcal{L}, \forall d, \forall h \quad (52)$$

$$v_i^{t,d,h} - v_j^{t,d,h} = (r_{ij} f p_{ij}^{t,d,h} + x_{ij} f q_{ij}^{t,d,h}) / U_0, \quad \forall (i, j) \in \mathcal{L}, \forall d, \forall h \quad (53)$$

$$\left(U_i^{\min}\right)^2 \leq v_i^{t,d,h} \leq \left(U_i^{\max}\right)^2, \quad \forall i \in \mathcal{N}, \forall d, \forall h \quad (54)$$

$$v_0^{t,d,h} = U_0^2, \quad \forall d, \forall h \quad (55)$$

$$\left(p_{mv}^{t,d,h}\right)^2 + \left(q_{mv}^{t,d,h}\right)^2 \leq \left(S_{mv}^{\max}\right)^2, \quad \forall d, \forall h \quad (56)$$

2.3. Compact matrix form

For season t , let $\mathbf{x}_t$ represent state variables (i.e., the variables linking adjacent seasons) and $\mathbf{y}_t$ be local variables (i.e., the variables only corresponding to season t). In the proposed formulation, $\mathbf{g}_{shs,i}^{t,end,H+1}$ serves as the state variable and the others are all local variables. Then, the NHOP problem mentioned in (1)–(56) can be mathematically stated in a compact form as:

$$\Phi = \min \sum_{t=1}^T \mathbf{d}_t^T \mathbf{y}_t \quad (57)$$

s.t. $(\mathbf{x}_t, \mathbf{y}_t) \in \mathcal{X}_t(\mathbf{x}_{t-1}), \quad t = 1, \dots, T$

$$\mathbf{x}_t \in \mathbb{R}^{x_1}, \mathbf{y}_t \in \mathbb{R}^{y_1} \times (0, 1)^{y_2}, \quad t = 1, \dots, T$$

where $\mathcal{X}_t(\mathbf{x}_{t-1})$ denotes the constraint set of season t . Formulation (57) is a classic multistage program, in which each season t corresponds to a stage.

2.4. Multi-stage stochastic program extension

The NHOP formulation in (57) is a deterministic optimization problem. However, the NHOP decisions could be significantly affected by various supply–demand uncertainties, i.e., the seasonal large-scale fluctuation and hourly small-scale variation of RES outputs and energy demands. Hence, we extend the deterministic formulation to a stochastic one for handling the complex uncertainties.

A stochastic process $\{\tilde{\xi}_t\}_{t=1,\dots,T}$, which reveals gradually over stages, is defined to capture these supply–demand uncertainties. $\tilde{\xi}_t$ denotes the uncertain factors of stage t , and is assumed to be stage-wise independent. Then, we reformulate the deterministic NHOP in (57) as the following multistage stochastic mixed-integer program (MS-MIP):

$$\Phi = \min_{\mathbf{x}_1, \mathbf{y}_1 \in \mathcal{X}_1} \mathbf{d}_1^T \mathbf{y}_1 + \mathbb{E}_{\tilde{\xi}_2} \left[\min_{\mathbf{x}_2, \mathbf{y}_2 \in \mathcal{X}_2(\mathbf{x}_1, \tilde{\xi}_2)} \mathbf{d}_2^T \mathbf{y}_2 \right. \\ \left. + \mathbb{E}_{\tilde{\xi}_3} \left[\dots + \mathbb{E}_{\tilde{\xi}_T} \left[\min_{\mathbf{x}_T, \mathbf{y}_T \in \mathcal{X}_T(\mathbf{x}_{T-1}, \tilde{\xi}_T)} \mathbf{d}_T^T \mathbf{y}_T \right] \dots \right] \right] \quad (58)$$

where $\mathbb{E}_{\tilde{\xi}_t}[\cdot]$ denotes the expectation with respect to a specific probability distribution of $\tilde{\xi}_t$. Note that (58) has a nested structure, reflecting the “non-anticipativity”, i.e., the decisions on stage t should not depend on the realizations of uncertainties in the future [29].

3. Solution methodology

3.1. Discretization through scenario tree

To address the complicated nested structure in (58), we first discretize the stochastic process to form a scenario tree, as shown in Fig. 2(a). In stage t , each node $\tilde{\xi}_t^\ell$ represents the ℓ -th actual realization of $\tilde{\xi}_t$. It is connected to every node in stage $t+1$ because of the stage-wise independence. We assume the number of actual realizations in each stage to be equal, denoted by $|\Xi|$. So the scenario tree in Fig. 2(a) can be further recombined to Fig. 2(b), where each path from the root node to one end is defined as a “scenario”. The total number of scenarios is $|\Xi|^{T-1}$, which grows with $|\Xi|$ in $(T-1)$ power. Besides, ξ_t^m denotes the realization of m th scenario in stage t .

3.2. Nested decomposition algorithm

An intuitive way to deal with (58) is to directly solve its extensive form [29]. Nevertheless, it may become computational intractable when $|\Xi|$ enlarges, even using the state-of-the-art optimization solver, e.g., Gurobi. To efficiently address the large-scale MS-MIP, we customize a nested decomposition algorithm based on SDDiP.

3.2.1. DP reformulation

Before introducing SDDiP, we equivalently reformulate (58) by a set of dynamic programming (DP) equations:

$$\Phi = \min_{\mathbf{x}_1, \mathbf{y}_1 \in \mathcal{X}_1} \mathbf{d}_1^T \mathbf{y}_1 + \mathbb{E}_{\tilde{\xi}_2} \left[Q_2(\mathbf{x}_1; \tilde{\xi}_2) \right] \quad (59)$$

for $t = 2, \dots, T$,

$$Q_t(\mathbf{x}_{t-1}; \tilde{\xi}_t) = \min_{\mathbf{x}_t, \mathbf{y}_t \in \mathcal{X}_t(\mathbf{x}_{t-1}, \tilde{\xi}_t)} \mathbf{d}_t^T \mathbf{y}_t + \mathbb{E}_{\tilde{\xi}_{t+1}} \left[Q_{t+1}(\mathbf{x}_t; \tilde{\xi}_{t+1}) \right] \quad (60)$$

where $Q_t(\mathbf{x}_{t-1})$ is the cost-to-go function. The expected cost-to-go (ECTG) function can be defined as:

$$Q_t(\mathbf{x}_t) := \mathbb{E}_{\tilde{\xi}_{t+1}} \left[Q_{t+1}(\mathbf{x}_t; \tilde{\xi}_{t+1}) \right] \quad (61)$$

For $t = T$, $Q_T(\mathbf{x}_{T-1}) = 0$.

3.2.2. Stochastic dual dynamic integer programming

The basic idea of SDDiP is to decompose the DP reformulation (59)–(60) by stages through constructing an under-approximation for the ECTG function. Then, it generates cutting planes through forward passes and backward passes alternatively to strengthen this approximation, until the updated approximation is “tight” or good enough.

Specifically, for each iteration τ , SDDiP randomly samples M scenarios from the scenario tree. Then, a forward pass starts on the M selected scenarios, from $t = 1$ to T , computing the stage-wise forward subproblem $\mathcal{P}_t^\tau(\hat{\mathbf{x}}_{t-1}^{m,\tau}, \xi_t^m)$ ($m = 1, \dots, M$), which can be expressed as:

$$Q_{-t}^\tau(\hat{\mathbf{x}}_{t-1}^{m,\tau}, \xi_t^m) := \min_{\mathbf{x}_t, \mathbf{y}_t} (\mathbf{d}_t^m)^T \mathbf{y}_t + \Psi_t \quad (62)$$

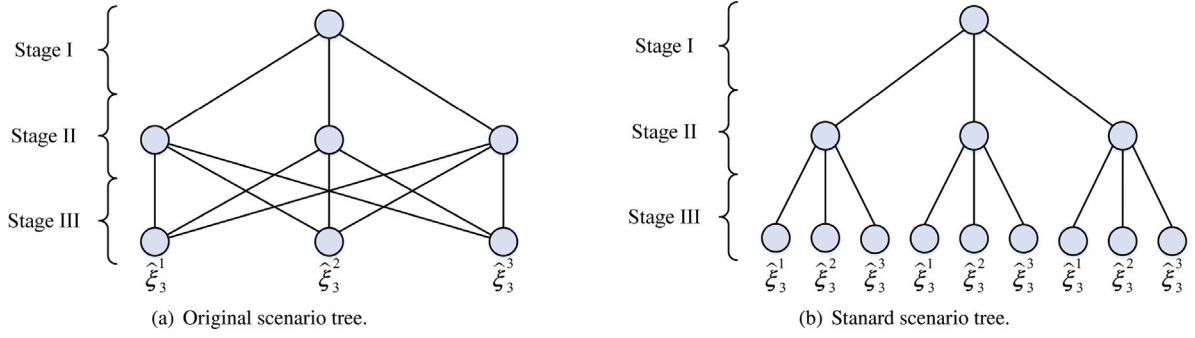
$$\text{s.t. } (\mathbf{x}_t, \mathbf{y}_t, \mathbf{z}_t) \in \mathcal{X}_t(\xi_t^m) \quad (63)$$

$$\mathbf{z}_t = \hat{\mathbf{x}}_{t-1}^{m,\tau} \leftarrow \boldsymbol{\pi}_t \in \mathbb{R}^{x_2} \quad (64)$$

$$\Psi_t \geq \frac{1}{|\Xi|} \sum_{\ell=1}^{|\Xi|} \left[\hat{\mathbf{v}}_t^{p^\ell} + \left(\hat{\boldsymbol{\pi}}_t^{p^\ell} \right)^T \mathbf{x}_t \right], \quad \forall p \in \mathcal{P}_t^\tau \quad (65)$$

$$\mathbf{x}_t \in \{0, 1\}^{x_2}, \mathbf{y}_t \in \{0, 1\}^{y_1} \times \mathbb{R}^{y_2}, \mathbf{z}_t \in [0, 1]^{x_2} \quad (66)$$

where Ψ_t is the approximated ECTG function, which is constructed by a set of hyperplanes in (65). $\mathbf{z}_t$ equals to the optimal $\hat{\mathbf{x}}_{t-1}^{m,\tau}$ derived from the forward problem of scenario m in stage $t-1$, and (64) is introduced for cut generation in the backward pass. $\boldsymbol{\pi}_t$ is the Lagrangian multiplier of (64). Besides, $p/\mathcal{P}_t^\tau$ is the index/set of cutting planes, while $\hat{\mathbf{v}}_t^{p^\ell}$ and $\hat{\boldsymbol{\pi}}_t^{p^\ell}$ represent two parameters that define the cuts. Moreover, $\mathcal{X}_t'$ is utilized

Fig. 2. Description of scenario tree ($|\Xi| = 3$).

to denote the combined feasible region surrounded by (63), (65) and (66).

Remark 2. Note that SDDiP requires all state variables for the MS-MIP be binary [30]. To this end, we approximately discretize the state variables ($g_{shs,i}^{t,end,H+1}$) to integer and apply the binarization technique.

Subsequently, a backward pass, proceeding from $t = T$ to 2, is conducted to generate valid cutting planes to strengthen the approximation Ψ_{t-1} of stage $t - 1$. To balance the solution quality and computation time, the Strengthened Benders (SB) cut [30] is adopted here as the form of cutting planes. Detailed steps of yielding SB cuts are presented as follows:

- STEP 1. In each stage t , for sampled node m , $|\Xi|$ LP relaxation of $\mathcal{P}_t^{m,\tau}(\hat{x}_{t-1}^{m,\tau}, \hat{\xi}_t^\ell)$ ($\ell = 1, \dots, |\Xi|$) are computed to obtain the multipliers $\hat{\pi}_{t,LP}^{m,\tau}$ of (64).
- STEP 2. By relaxing (64) and setting π_t to $\hat{\pi}_{t,LP}^{m,\tau}$, we solve $|\Xi|$ Lagrangian relaxation (LR) problems of $\mathcal{P}_t^{m,\tau}(\hat{x}_{t-1}^{m,\tau}, \hat{\xi}_t^\ell)$ in (67) to attain the optimal values ($\hat{Q}_{t,LR}^{m,\tau}$) of its objective function.

$$\begin{aligned} \hat{Q}_{t,LR}^{m,\tau}(\hat{x}_{t-1}^{m,\tau}, \hat{\xi}_t^\ell) = \\ \min_{(x_t, y_t, z_t, \Psi_t) \in \mathcal{X}_t'(\hat{\xi}_t^\ell)} (d_t^m)^T y_t + \Psi_t + \left(\hat{\pi}_{t,LP}^{m,\tau} \right)^T (\hat{x}_{t-1}^{m,\tau} - z_t) \end{aligned} \quad (67)$$

- STEP 3. Generate the SB cuts as below:

$$\Psi_{t-1} \geq \frac{1}{|\Xi|} \sum_{\ell=1}^{|\Xi|} \left[\hat{Q}_{t,LR}^{m,\tau} + \left(\hat{\pi}_{t,LP}^{m,\tau} \right)^T (x_{t-1} - \hat{x}_{t-1}^{m,\tau}) \right], \quad m = 1, \dots, M \quad (68)$$

Then, by collecting M cuts from stage t and adding them to the cut set in stage $t - 1$, it can be updated as $\mathcal{P}_{t-1}^{\tau+1}$.

When finishing the backward pass, the relaxation bound (RB) can be calculated by solving $\mathcal{P}_1^{\tau+1}$. The SDDiP algorithm terminates when RB becomes stable [30]. After that, a feasible bound (FB) can be estimated through the sampling of a large scenario subset out of the scenario tree, and conducting a complementary forward pass.

3.2.3. Solution procedures

Based on our elaboration for forward pass, backward pass and strengthened cut set, the complete procedures to implement SDDiP are summarized as in Algorithm 1.

Algorithm 1 Stochastic dual dynamic integer programming

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1: Initialization: Set  $RB \leftarrow -\infty$ ; Initialize  $\{\mathcal{P}_t^1\}_{t=1,\dots,T}$ 
2: while  $RB$  has not became stable do
3:   Sample  $M$  scenarios  $\{\xi_1^m, \dots, \xi_T^m\}_{m=1,\dots,M}$  from the scenario tree.
4:   Forward Pass:
5:   for  $m = 1, \dots, M$  do
6:     for  $t = 1, \dots, T$  do
7:       Solve forward subproblem  $\mathcal{P}_t^{m,\tau}(\hat{x}_{t-1}^{m,\tau}, \xi_t^m)$ ;
8:       Collect optimal  $\hat{x}_t^{m,\tau}$  and  $\hat{y}_t^{m,\tau}$ .
9:     end for
10:    Calculate policy value  $\bar{\Phi}^{m,\tau} := \sum_{t=1}^T (d_t^m)^T \hat{y}_t^{m,\tau}$ .
11:   end for
12:   Backward Pass:
13:   for  $t = T, \dots, 2$  do
14:     for  $m = 1, \dots, M$  do
15:       for  $\ell = 1, \dots, |\Xi|$  do
16:         Solve LP relaxation of  $\mathcal{P}_t^{m,\tau}(\hat{x}_{t-1}^{m,\tau}, \hat{\xi}_t^\ell)$ ; Collect  $\hat{\pi}_{t,LP}^{m,\tau}$ .
17:         Fix  $\pi_t = \hat{\pi}_{t,LP}^{m,\tau}$  and solve LR problem in (67); Collect  $\hat{Q}_{t,LR}^{m,\tau}$ .
18:       end for
19:       Generate an SB cut as (68).
20:     end for
21:   end for
22:   EB Update:
23:   Update EB by solving  $\mathcal{P}_1^{\tau+1}$ .
24:    $\tau \leftarrow \tau + 1$ .
25: end while
26: Feasible Bound Acquisition:
27: Resample  $M'$  scenarios; Carry on a Forward Pass on them.
28: Obtain expectation  $\mu = \frac{1}{M'} \sum_{m=1}^{M'} \bar{\Phi}^{m,\tau}$  and standard deviation  $\sigma = \sqrt{\frac{1}{M'-1} \sum_{m=1}^{M'} (\bar{\Phi}^{m,\tau} - \mu)^2}$  of the  $M$  policy values:
29: Obtain  $FB = \mu + \frac{1.96\sigma}{\sqrt{M}}$ .

```

Remark 3. Note that for the resample step in Algorithm 1, the selection strategy for M' is case by case. When the scale of entire scenario set is small, M' can even be set to $|\Xi|^{T-1}$, so that μ itself serves as the FB. However, if $|\Xi|^{T-1}$ is large, it could be very time-consuming to compute the forward subproblems on all scenarios. In these instances, M' would be a relatively small value (but still much bigger than M).

4. Case studies

The proposed NHOP model and customized SDDiP algorithm are verified based on an exemplary 33-bus integrated energy network with 3 HMMs and 2 RES-based microgrids (RMs), as shown in Fig. 3. Table 1 exhibits the capacity sizing schemes of all these microgrids. Major techno-economical parameters for our operational planning model are listed in Table 2. The time-of-use (TOU) electricity prices are specified as in Table 3. Besides, we consider a one-year operation horizon

Table 1
Capacity configuration of HMMs and RMs.

Microgrid	Bus	PV (kW)	WT (kW)	BS (kW)	BS (kWh)	ELZ (kW)	HS (kg)	SHS (kg)	FC (kW)	HWS (kWh)	CWS (kWh)	AC (kWh)
HMM1	2	5000	4000	2000	4000	2000	200	8000	1000	1500	1200	1000
HMM2	13	4000	5000	1500	3000	2000	150	5000	1000	1000	600	800
HMM3	27	3000	3500	1200	3000	2000	100	4000	1000	1000	600	800
RM1	24	4000	5000	1500	3000	0	0	0	0	0	0	0
RM2	31	3000	4000	1000	3000	0	0	0	0	0	0	0

Table 2
Major techno-economical parameters for NHOP [31].

Parameter	Description	Value
$\zeta_{bs}/\zeta_{elz}/\zeta_{fc}$	Degradation cost factor of BS/ELZ/FC (\$/kWh)	0.001
κ_{ec}	Penalty cost factor of electricity curtailment (\$/kWh)	20
$\iota_{es}/\iota_{hs}/\iota_{cs}$	Penalty cost factor of electrical/heating/cooling load shedding (\$/kWh)	20
χ_{e-g}	COP of ELZ for electricity-hydrogen conversion (kg/kWh)	0.022
χ_{g-e}	COP of FC for hydrogen-electricity conversion (kWh/kg)	9.09
χ_{g-h}	COP of FC for hydrogen-heating conversion (kWh/kg)	19.91
χ_{h-e}	COP of AC for heating-cooling conversion	0.94

Table 3
TOU electricity price.

Period (h)	1–8	9–10	11–14	15–17	18	19–22	23	24
$\varrho_{imp}^{t,d,h}$ (\$/kWh)	0.0431	0.1135	0.1875	0.1135	0.1875	0.2058	0.1875	0.1140

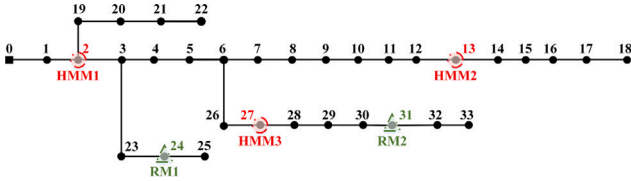


Fig. 3. 33-bus test network with HMMs and RMs.

including four interdependent stages ($T = 4$), which corresponds to four seasons in order.

Fig. 4 presents the typical varying patterns of RES outputs and multi-energy demands. The RES outputs and electrical demands are extracted from the operation data of real-world power grids in Northern China, while heating and cooling demands are simulated using Energy-Plus [32]. As in Fig. 4(a), the PV outputs demonstrate evident diurnal and seasonal rhythms, which are consistent with the variation trends of solar irradiation. On the one hand, the PV outputs climb up from morning to the noontime, reaching the climax at around 13:00 and monotonously decrease during the afternoon. On the other hand, the average PV outputs could be higher in a Summer day than Spring and Autumn, while reaching the lowest level in Winter. In contrast, the fluctuation of WT outputs (as in Fig. 4(b)) is more uncertain, which could be higher at a Spring day than the rest seasons. Also, its diurnal peak might appear in different time periods. As in Fig. 4(c), the daily variation of electrical demands can be described as “camel” curves, while their seasonal rhythm is less obvious. In addition, for typical residential or commercial areas, the heating demands mainly occur in Winter and reach a peak at early morning, while the cooling demands arise in Summer and reach a peak during the afternoon, as can be seen in Fig. 4(d).

The numerical experiments are performed under the environment of MATLAB R2023b on a mobile workstation with Intel(R) Core(TM) i9-13900H processor and 16 GB RAM. In SDDiP procedure, we set $M = 1$. All the basic MIP and LP problems are solved using YALMIP Toolbox [33] and Gurobi 11.0 (a state-of-the-art commercial solver) [34]. The solution gap of Gurobi keeps its default value. SDDiP terminates when RB becomes stable, and the time limit is set to 3600 s. Then, a forward pass is conducted complementally upon $\min\{|\Xi|^{T-1}, 125\}$ scenarios to calculate FB.

4.1. Analysis of long-term operation results

We first present the long-term operation scheme of HMMs under an average scenario to investigate the synergy of multi-type energy storage devices. Figs. 5–7 show the NHOP results of electrical, hydrogen, as well as heating and cooling subsystems, respectively. Each figure contains two or four typical days corresponding to different seasons, which captures the seasonal and diurnal patterns of system operation. Our results are elaborated and discussed as in the following:

- **Electrical Subsystem:** According to Fig. 5, the operational decisions of electrical subsystem are made based on the varying patterns of RES outputs, electrical demands, and TOU electricity prices. As shown in Fig. 5(a), because the wind power resources are obviously more abundant in Spring than the other seasons while the demands level is relatively lower, the electrical demands in a typical Spring day can be mostly fulfilled by RES outputs. Moreover, during 8:00–18:00, since the summation of WT and PV generation exceeds the load demand, both the BSs and ELZs are activated to store the surplus energy. For such a Spring scenario, the regional electrical subsystem can autonomously operate without the support of utility power grid. The operation strategies in Autumn day (as in Fig. 5(c)) are similar to the Spring, except the needs for grid power imports and the less frequent utilization of energy storage devices. That is because the WT outputs are much lower along with a higher electrical demands level in Autumn. In comparison, the situations in Summer and Winter could be quite different, associating with the variation of heating and cooling demands. As exhibited in Fig. 5(b), for a typical Summer day, our optimization results suggest a mixed power supply scheme based on the generation of WTs, PVs and FCs. It mainly relies on utility power procurement during early morning (e.g., 1:00–7:00) to energize the electrolyzers under the valley of TOU prices, in preparation for satisfying the power and cooling demands in subsequent periods. The power import also occurs in night time (e.g., 20:00–24:00) when PVs’ generation is unavailable. Similarly, in Winter (see Fig. 5(d)), it intends to purchase power from the utility grid in the morning time but with a smaller amount, because the WTs’ generation is more abundant than Summer. Besides, the FCs operate under a relatively high setting point during 5:00–10:00 following the peak periods of heating demands.

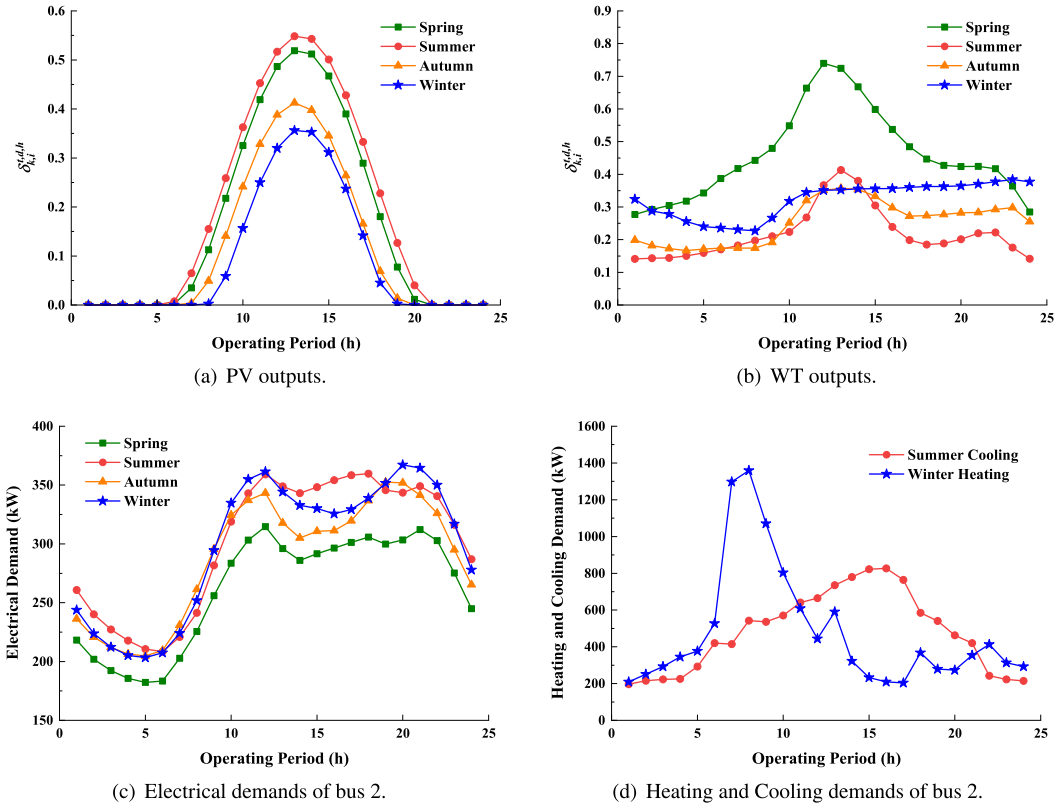


Fig. 4. Typical seasonal and daily varying patterns of RES outputs and multi-energy demands.

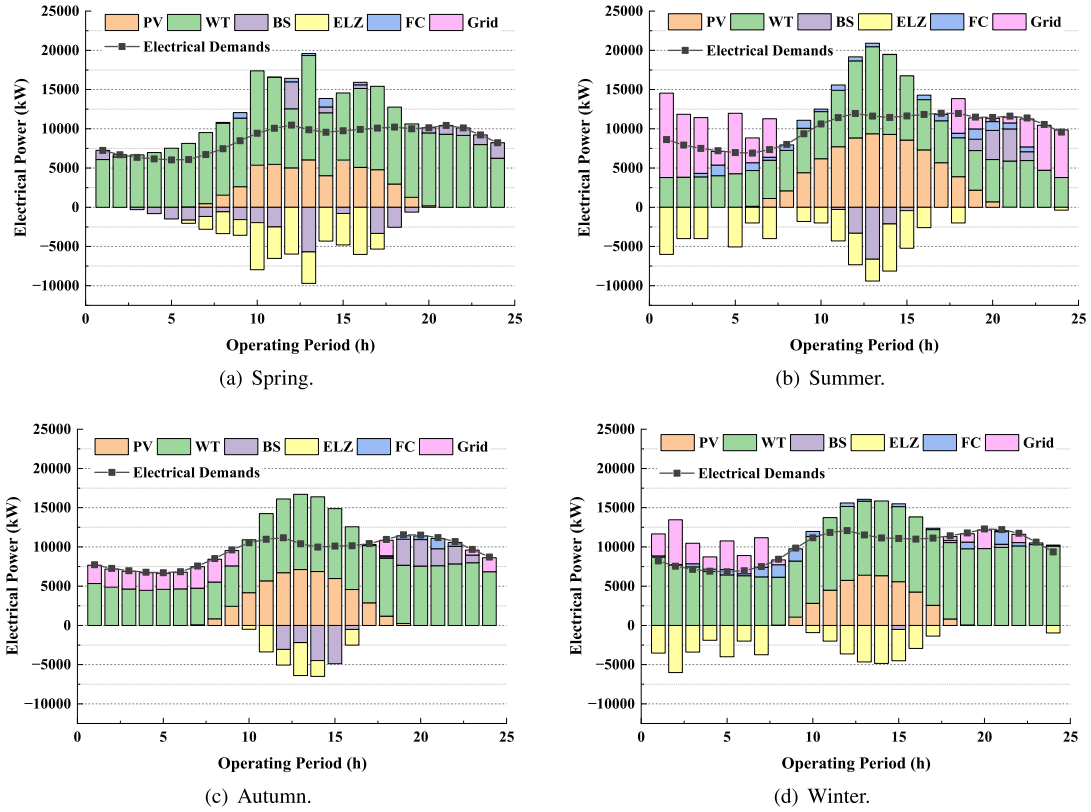


Fig. 5. Seasonal operation scheme of electrical subsystem.

• *Hydrogen Subsystem:* The hydrogen subsystem in an HMM is responsible for large-scale and long-term energy storage, also serves

as the critical energy resource for supplying heating and cooling demands through the H-CCHP system. As shown in Figs. 6(a) and

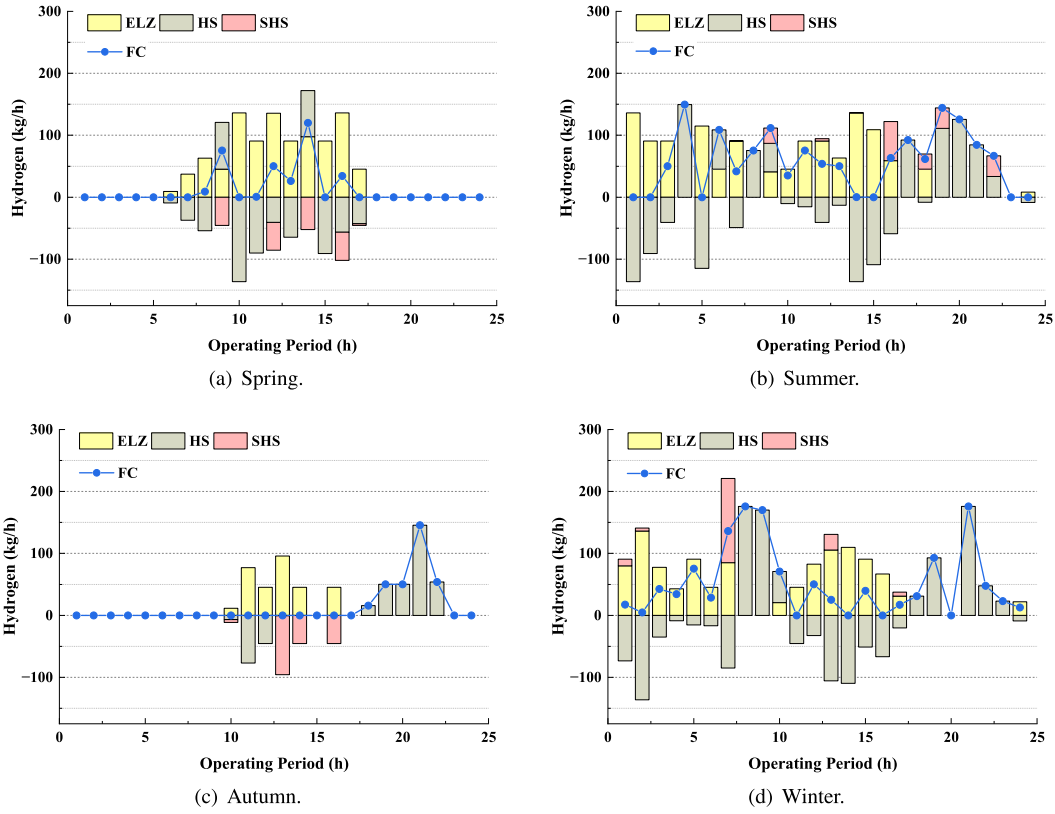


Fig. 6. Seasonal operation scheme of hydrogen subsystem.

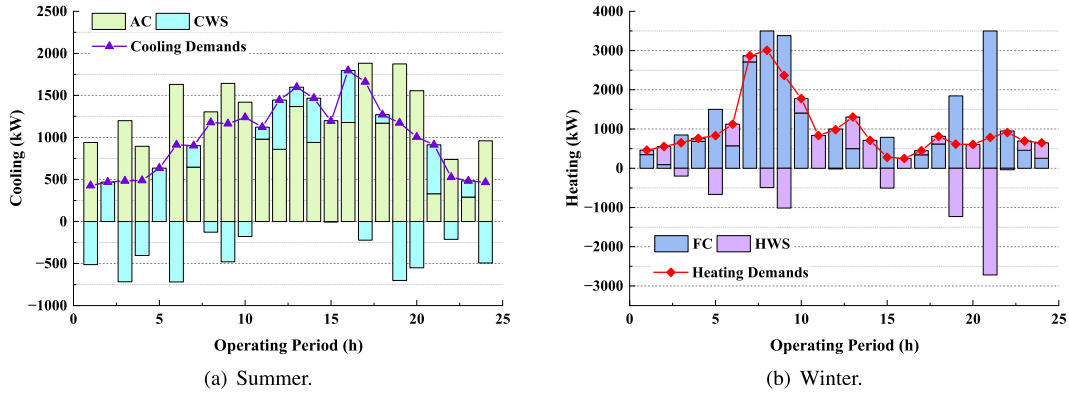


Fig. 7. Seasonal operation scheme of thermal supply subsystem.

6(c), the ELZs, HSs and SHSs mainly work during the day time (e.g., 8:00–16:00) in Spring and Autumn for accommodating the excessive power of RES units. As a contrast, in Summer or Winter (see Figs. 6(b) and 6(d)), the hydrogen storage system starts up for an entire day to keep track with the demands pattern of cooling and heating consumption.

- **Thermal Supply Subsystem:** Note that the cooling demands are only considered in Summer day, as in Fig. 7(a). It can be observed that ACs, which operate based on FCs' generation, nearly remain active for a whole day to fulfill the cooling demands. Similarly, as exhibited in Fig. 7(b), the heating demands in Winter days are mainly served by FCs. Moreover, the heating and cooling storage facilities are activated for further enhancing the operational flexibility of H-CCHP system.

Specifically, we study the long-term storage dynamics of SHSs. As shown in Fig. 8, the operational decisions of SHS demonstrate a clear

seasonal pattern, i.e., charging in Spring and Autumn, while discharging in Summer and Winter. Together with the results in Figs. 4 and 6, we can see that the charging of SHS in transition seasons is not only for renewable energy utilization, but also to prepare for the upcoming requirements of large-scale thermal supply. From Spring to Summer, there also exists a significant increase in electrical loads. Along with the cooling demands, the SHS begins to release hydrogen for energizing the co-generation system. Besides, due to inadequacy of RES outputs in Winter, the SHS also operates under the energy discharging mode that facilitates the coordinated heat and power supply. Therefore, our results confirm the validity of SHSs for inter-seasonal energy regulation.

4.2. Impact of renewable energy penetration

Based on the case settings in Section 4.1, the influence of RES penetration on the operational economics and efficiency of multi-microgrid network is further explored. The capacity factor γ_{res} is introduced

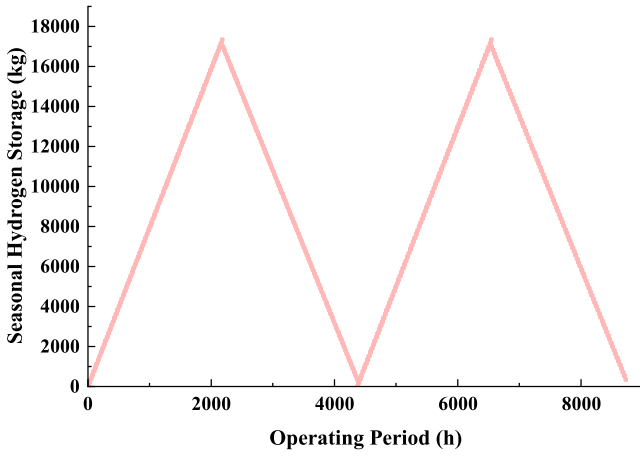


Fig. 8. Yearly storage states of SHS.

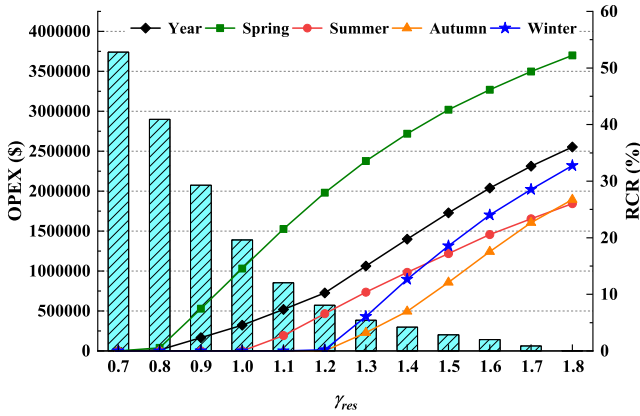


Fig. 9. Economic and energy efficiency metrics with changing capacity of RES integration.

to characterize the proportional changes of total RES capacity. With respect to a varying γ_{res} , we present the values of OPEX and RES curtailment rate (RCR) in Fig. 9. In this case, the penalty costs are not considered in the evaluation of OPEXs. It can be seen that the OPEX decreases monotonously (from \$3,740,832.09 to 0) with the increasing of γ_{res} (from 0.7 to 1.8). For a higher capacity of RES units, it needs less importing power through the utility grid, thus reducing the power procurement costs. When $\gamma_{res} = 1.8$, the cost of this part reduces to zero. The remaining OPEX is rather small (i.e., \$4622.55), which only constitutes the operational costs of BSs, ELZs and FCs.

On the other hand, the increase of γ_{res} would lead to a growing trend of RCR. We observe a more drastic increase of RCR under Spring scenario than the rest seasons and yearly average, which is consistent with the seasonal patterns of WT and PV outputs. When γ_{res} increases from 0.7 to 1.8, the RCR in Spring grows sharply from 0 to 52.23%. Differently, the changing of RCR in Autumn could be slower and more smooth, which, again, reflects the insufficiency of wind and solar resources in this season. Also, the RCR levels in Summer and Winter are generally higher than Autumn. It attributes to both the seasonal resource patterns and the influence of thermal supply. To ensure the reliable supply of cooling or heating demands, sometimes we may need FCs to generate more rather than fully utilizing the RES generation. So, the proposed NHOP method can help determine the proper ratios of energy curtailment under different season scenarios and RES penetrations.

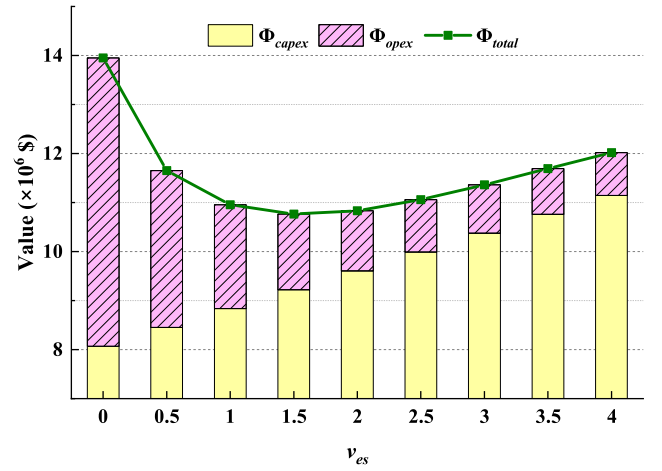


Fig. 10. Cost-benefit metrics with changing installed capacity of energy storage.

4.3. Cost-benefit analysis for different storage sizing plans

In this subsection, we make use of the NHOP model to conduct the cost-benefit analysis for different configuration plans of energy storage. Another capacity factor v_{es} is introduced to quantify the proportional changes of storage capacity all over the multi-microgrid network. In Fig. 10, the annual capital expenditure of storage configuration (Φ_{capex}) is compared with the OPEX (Φ_{opex}) under different levels of v_{es} . Note that the penalty factors are considered and set to 0.2 when calculating Φ_{opex} . The capital cost Φ_{capex} can be attained according to Refs. [11,23,35]. The total system costs (i.e., $\Phi_{total} = \Phi_{capex} + \Phi_{opex}$) are also depicted in Fig. 10 to assess the overall economical effectiveness.

With the enlarging of v_{es} , we observe a monotonous increasing trend of Φ_{capex} along with the decline of Φ_{opex} . As expected, Φ_{capex} grows linearly from \$8,068,699.60 to \$11,144,040.29. In contrast, the changing rate of Φ_{opex} may experience different phases. When v_{es} changes from 0 to 2, the operational cost drops very fast (decreasing by 79.14%). It could be induced by the dramatic declining of RES curtailment cost with more storage deployed. When v_{es} changes from 2 to 4, the decreasing of Φ_{opex} slows down (reduces by 28.79%). In this phase, the decrease of OPEX mainly originates from the energy arbitrary of storage facilities, which is with a smaller unit cost than the curtailment penalties. As a summation of Φ_{capex} and Φ_{opex} , Φ_{total} first decreases when v_{es} changes in the interval of [0, 1.5], then increases when v_{es} changes between [1.5, 4.0]. The lowest cost level (\$10,764,844.83) can be achieved when $v_{es} = 1.5$, which corresponds to a best trade-off between investment and operational expenditures. Practically, the results in Fig. 10 could guide the system planners to select the optimal configuration of multi-type energy storage.

The cost-benefit analysis is further implemented for different types of energy storage (including the batteries, hot and chilled water tanks, as well as the short-term and seasonal hydrogen storage), as presented in Fig. 11. The individual capacity factors v_k are applied. As can be seen in Fig. 11(a), the total cost of BS decrease when v_{bs} changes from 0 to 1, but monotonously grows subsequently. It indicates that the high capital cost of BSs (mainly due to the capacity degradation) overweighs their effects on RES utilization and energy arbitrary. Following this result, a large capacity of BS installation may not be recommended. As in Figs. 11(b)–11(d), the total cost of other four storage types decreases monotonously with the increasing of v_k . In particular, the cost drop of HS is very sharp at the beginning while slows down when $v_{HS} \geq 1.5$. After that, the change of v_{HS} could barely influence their cost variation. In comparison, the cost of seasonal storage system keeps declining at a fair rate, which demonstrates a more flexible design space for SHS deployment to improve the cost-efficiency.

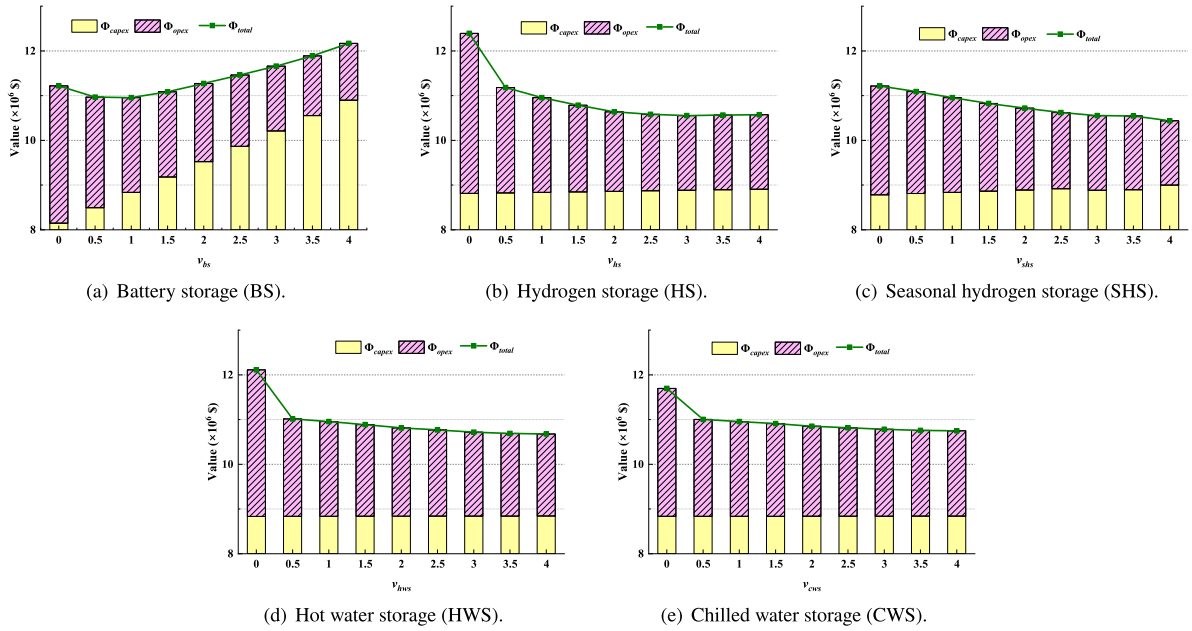


Fig. 11. Detailed cost-benefit analysis for different storage types.

Table 4

Value of stochastic solutions under different scenario trees.

$ \Xi $	$ \Xi ^{T-1}$	Φ_S (\$)	Φ_D (\$)	VSS
2	8	7 131 591	7 314 917	2.51%
3	27	5 430 004	5 636 096	3.66%
4	64	6 251 382	6 367 591	1.83%

These observations verify the effectiveness of our NHOP model for evaluating the cost-benefits of different storage technologies and their capacity sizing schemes, which thus could provide informative planning guidance for the HMMs.

4.4. Value of uncertainty modeling

In this subsection, we further verify the effectiveness of developing the MS-MIP extension to handle time-dependent supply-demand uncertainties. In particular, the value of stochastic solution (VSS) index is introduced to measure the cost-saving effects of our uncertainty modeling:

$$VSS = \frac{\Phi_D - \Phi_S}{\Phi_D} \times 100\% \quad (69)$$

where Φ_S denotes the objective value of stochastic model in (58). Also, by fixing the state variables in (58) as the solution of deterministic model (57), we can derive Φ_D by resolving problem (58). Table 4 presents the VSS metrics based on the operational results with 8, 27 and 64 scenarios in a scenario tree. It can be seen that VSS keeps positive in all the instances with an average value of 2.67%, demonstrating that our MS-MIP formulation can effectively decrease the risk cost of microgrids operation.

The instance with $|\Xi| = 4$ is utilized to elaborate the NHOP results with stochastic modeling. The stage-wise (seasonal) realizations of four selected scenarios are given in Table 5. For simplicity, we choose two parameters κ_{res}^t and κ_{demand}^t respectively representing the realization of RES generation and multi-energy demands (i.e., power, heating and cooling) of each season compared to the deterministic case. Fig. 12 shows the SOC of SHS at the end of each season under these scenarios. As the starting stage, the SOC of SHSs keeps the same for all the Spring scenarios due to the non-anticipative requirements for multi-stage operational decision. For the subsequent stages, the varying of

Table 5

Seasonal realizations of RES generation and energy demands.

Scenario		Spring	Summer	Autumn	Winter
A	κ_{res}^t	1	0.8	1.4	0.9
	κ_{demand}^t	1	1.4	0.6	0.9
B	κ_{res}^t	1	0.8	1.4	1.2
	κ_{demand}^t	1	1.4	0.6	0.8
C	κ_{res}^t	1	1.4	1.2	1.4
	κ_{demand}^t	1	0.6	0.8	0.6
D	κ_{res}^t	1	1.2	0.9	1.2
	κ_{demand}^t	1	0.8	0.9	0.8

SOCs largely depends on the realization of uncertainties. For instance, compared to Scenario A, Scenario B is with higher level of RES outputs and lower energy demands in Winter. As a result, the SHS in Scenario A releases more hydrogen than in Scenario B. Such supply-demand gap is enlarged in Scenario C, which further restricts the utilization of SHS. In comparison to Scenario B, the final remaining hydrogen storage increases from 5868 kg to 17,000 kg. Hence, different from the deterministic model, the results of MS-MIP could assist the microgrids operators to flexibly adjust their operation and risk-hedging strategy against versatile practical situations.

4.5. Computational tests

The computational feasibility of using SDDiP to solve our stochastic formulation is examined and reported in 6. Note that a parallelization strategy is leveraged in the forward and backward passes to enhance the solution performance. For comparison, the results derived by the direct use of Gurobi (a state-of-the-art commercial solver) are also given. As in Table 6, Column 1 and 2 show the stage number ($T = 4$ in all instances), while Column 3 gives the total number of scenarios. Column 4–8 presents the performance metrics of SDDiP, including FB, RB, remaining gap, number of iterations and the solution time. Column 9–11 provides the metrics of Gurobi, including FB, remaining gap and the solution time. Besides, all the algorithm implementations will be restricted by a time limit as 3600 seconds (s). By sharing the same RB

Table 6
Results of computational tests.

T	Ξ	Ξ ^{T-1}	SDDiP					Gurobi		
			FB	RB	Gap	Iter	Time (s)	FB	Gap	Time (s)
4	1	1	617 826.61	616 538.76	0.21%	5	29.84	617 824.39	0.21%	6.41
	2	8	807 960.90	807 966.67	0.00%	8	54.38	807 959.35	0.00%	26.71
	3	27	610 803.16	610 767.42	0.01%	8	67.28	610 788.18	0.00%	151.87
	4	64	559 546.70	559 545.09	0.00%	10	96.74	559 546.19	0.00%	659.16
	5	125	566 890.74	566 547.74	0.06%	13	140.29	566 703.09	0.03%	2272.5
	6	216	655 269.41	654 692.02	0.09%	11	156.08	\	\	>3600
	10	1000	566 183.39	566 173.96	0.00%	13	193.16	\	\	>3600

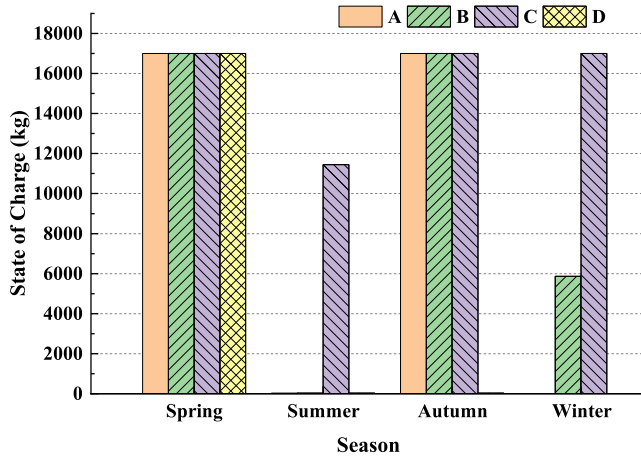


Fig. 12. Relationship between OPEX and RCR with RES installations.

of SDDiP, the solution gaps of SDDiP and Gurobi can both be defined as:

$$\text{Gap} = \frac{FB - RB}{RB} \times 100\%. \quad (70)$$

As shown in Table 6, SDDiP successfully deal with all instances, providing high-quality solutions in a gap of 0.22% with acceptable amount of time (less than 195 s). The solution time for the instance with 1 scenario is 29.84 s, whereas for the instance with 1000 scenarios is 193.16 s. The solution time only increases by 6.47 times when the size of scenario tree is enlarged by 1000 times. As a contrast, the solution time of Gurobi rises very fast with the growth of problem scale. Although Gurobi demonstrates a better performance for small-scale instances (when $|\Xi|^{T-1} = 1$ or 8), it is defeated by SDDiP when 27 or more stochastic scenarios are involved. In some instances, SDDiP could even reduce the computation time by more than 94%. When $|\Xi|^{T-1} \geq 216$, Gurobi is incapable to find a feasible solution in predefined time limit. In comparison, SDDiP can readily address a very challenging problem with 1000 scenarios within only 193.16 s. Therefore, the proposed SDDiP algorithm provides a powerful and scalable computational tool to deal with large-scale MS-MIPs.

5. Conclusion and discussions

In this paper, we propose an MS-MIP method to support the networked HMMs operation planning considering seasonal and short-term energy storage. To relieve the computational burden of computing large-scale MS-MIP, a nested decomposition algorithm based on SDDiP is customized and developed. Case studies on a 33-bus integrated energy network verify the effectiveness of the proposed NHOP model and SDDiP algorithm. Some key observations and insights are summarized as follows:

- The seasonal hydrogen storage provides a long-term and large-scale energy regulation capability. Our results suggest that the SHS

charges in transitions seasons (Spring and Autumn) for fully utilizing the RES generation, while prepares for the upcoming demands of centralized cooling and heating consumption in Summer and Winter. The coordination of SHS with short-term energy storage helps achieve a higher operational flexibility for mitigating the demand–supply mismatches in a district multi-energy network.

- The proposed operation planning method offers an analytical tool to generate economical operation strategies for networked HMMs. Moreover, it can provide an informative planning guidance for the capacity configuration of RES units and multi-type energy storage. Compared to conventional storage equipment, SHS could be with a more flexible design space for enhancing the system-level economic benefits.
- The multi-stage stochastic formulation could help the grid operators to flexibly adjust their operation strategies against versatile practical situations, which effectively decreases the risk cost. Also, the customized SDDiP algorithm provides a powerful and scalable computational tool to deal with large-scale MS-MIPs. Compared to a state-of-the-art commercial solver (i.e., Gurobi), our method has reduced the solution time by more than 94% when ≥ 125 scenarios are involved.

CRedit authorship contribution statement

Xunhang Sun: Writing – original draft, Software, Methodology. **Xiaoyu Cao:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Miaomiao Li:** Visualization, Investigation. **Qiaozhu Zhai:** Validation, Methodology. **Xiaohong Guan:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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